



Designing for EMC Compliance

Pass the chamber on the first try

Conducted • Radiated • Susceptibility • ESD

Audience: Intermediate Engineering Students

EMI sources • Mitigation • Pre-compliance • Lab cost



What is EMC?

EMI = emitted noise; EMC = compatibility

Emissions

Your product radiates RF that bothers others.

Susceptibility

External RF disturbs your product's operation.

ESD

Static discharge — human touches device.

Surge

Lightning strike on power line
→ kV transient.

EFT/Burst

Switching transients on nearby wiring.

Power dips

Brownouts, sags, interruptions.

EMC Standards Landscape

Different markets, different tests

Market	Standard	Body	Notes
India	BIS / WPC	Govt of India	ETA / TEC certs
EU (CE)	EN 55032 / 55035	CENELEC	Mandatory for sale
USA	FCC Part 15	FCC	Class A vs B
Automotive	CISPR 25 / 36	ISO + OEM	Tougher than consumer
Industrial	EN 61000-6-x	IEC	Less strict, higher robustness
Medical	IEC 60601-1-2	IEC	Safety-of-life criteria
Aerospace	DO-160	RTCA	Severe ESD/lightning

Common EMC Tests

What labs measure — and what causes failures

Conducted EMI

30 kHz - 30 MHz on power lines.

Radiated EMI

30 MHz - 6 GHz from antenna probes.

ESD

8 kV contact / 15 kV air on every accessible surface.

EFT / Burst

Switching pulses on power line.

Surge

Lightning-equivalent on power lines.

Power dips

Brownouts. Must restart cleanly.

Where EMI Comes From

Sources to consider during design

- ▶ Switching regulators — primary radiated source above 30 MHz
- ▶ Clock harmonics — strong tones at multiples of clock frequency
- ▶ Digital fast edges — broadband noise up to a few GHz
- ▶ USB / Ethernet / HDMI signalling — cable-conducted
- ▶ Motors and relays — broadband bursts during switching
- ▶ Cables outside the enclosure — antenna for internal noise
- ▶ Loop currents — area determines radiation efficiency

EMC-Friendly PCB Layout

The cheapest place to fix EMI is the PCB

- ▶ Continuous ground plane — every signal has a clean return
 - ▶ Short, small switching loops — minimise loop area
 - ▶ Decouple every IC with multi-value caps
 - ▶ Spread-spectrum clocks reduce single-tone peaks
 - ▶ Slow edge rates if speed allows (TTL series resistors)
 - ▶ Twisted pair or differential signalling for long traces
- ▶ Filter every cable entering/leaving the PCB
 - ▶ Common-mode chokes on power & data lines
 - ▶ TVS diodes on every external interface
 - ▶ Ferrite beads on power lines feeding noisy loads
 - ▶ Shielding can over noisy sections (SMPS, RF)
 - ▶ Connector shells grounded to chassis at the bracket

Mitigation Toolkit

Components that solve EMC problems

Decoupling cap

100 nF + 10 nF + 1 nF
at every IC power pin.

Ferrite bead

High-Z at MHz/GHz,
low-Z at DC.

Common-mode choke

Two windings, shared core.
Kills common-mode currents.

TVS diode

Clamps ESD < 1 ns.
Protects downstream ICs.

MOV / GDT

Surge protection on
AC inputs.

Pi filter

C + L + C on supply lines.
Cheap, effective.

ESD Protection

First contact failure — and how to prevent it

- ▶ Human body model: 8 kV contact / 15 kV air discharge
- ▶ Every accessible signal pin needs a TVS to chassis ground
- ▶ USB / Ethernet / HDMI need low-capacitance TVS (won't slow signals)
- ▶ Place TVS as close to the connector as possible
- ▶ Ground the connector shell directly to chassis (not signal ground)
- ▶ Coverlay the PCB edge near connectors — keep traces > 1 mm from edge
- ▶ ICs marked 'ESD-protected' have on-die diodes but only HBM, not system-level
- ▶ Test with an ESD gun before sending to compliance lab

Pre-Compliance Testing

Find failures before the expensive lab visit

Near-field probes

Probe set scans board
to find hot-spot emitters.

LISN + spectrum analyzer

Conducted emissions
down to 30 kHz.

Anechoic chamber kit

Tiny chamber + probe antenna
gives ballpark radiated.

ESD gun

₹40-80k consumer-grade.
Finds open weaknesses.

Surge generator

Lab time at universities
/ partner consultancies.

EMC consultant

Half-day visit < ₹20k.
Saves weeks of rework.

Cost & Time Budget

Real numbers for Indian start-ups

Item	Cost (₹)	Lead time	Notes
Pre-compliance scan	20-50k	2-3 days	Bangalore / Pune labs
Full BIS CE 55032	3-6 L	4-8 weeks	Multiple test articles
FCC Part 15 (USA)	5-10 L	6-12 weeks	Plus an FCC ID
CE-RED	4-10 L	8-12 weeks	Including report
Re-test after fail	1-3 L	+2-4 weeks	Common — budget for it
EMC consultant	10-30k/day	Same week	Pre-design review

Lesson

Common EMC Test Failures

What you'll fail on if you skip mitigation

- ▶ Radiated emissions at clock harmonics — long traces near edge
- ▶ Conducted emissions at SMPS switching frequency — missing input filter
- ▶ ESD: random reset of MCU — no TVS on connector pin
- ▶ EFT bursts during PSU tests — CMC missing on supply
- ▶ Surge: blown TVS but cap network too small — board still resets
- ▶ USB-conducted emissions — no CMC on data lines
- ▶ Cable acts as antenna — no ferrite + shield termination

Best Practices Recap

12 habits that pass on first try

- ▶ Continuous ground plane — non-negotiable
- ▶ Multi-value decoupling at every IC
- ▶ TVS on every external signal pin
- ▶ CMC on every cable bundle
- ▶ Ferrite beads on noisy power lines
- ▶ Connector shells to chassis ground
- ▶ Spread-spectrum clocks when possible
- ▶ Slow edges where speed isn't critical
- ▶ Shielding cans on RF & SMPS
- ▶ Pre-compliance scan before lab booking
- ▶ EMC consultant on the design review
- ▶ Budget for one re-test cycle (it happens)

Key Takeaways

Key takeaways from this lecture

EMI = your problem

Standards require it for sale.

Ground plane = 80% fix

Continuous return path.

Pre-compliance saves money

Catch failures cheaply.

ESD = every connector

Low-cap TVS, close to pin.

CMC = every cable

Common-mode kills compliance.

Plan budget early

Lab time + re-test cycle.

Recommended Reading

EMC resources for self-study

- ▶ Henry Ott — 'Electromagnetic Compatibility Engineering' (the EMC bible)
- ▶ Tim Williams — 'EMC for Product Designers'
- ▶ Mark Montrose — 'EMC Made Simple'
- ▶ Robert Feranec YouTube — EMC walk-throughs
- ▶ EMCFastPass YouTube — pre-compliance tutorials
- ▶ Würth Elektronik 'Trilogy of Magnetics' — free book on inductors and ferrites
- ▶ EM Test / Schaffner application notes
- ▶ FCC OET 64 lab procedures (free download)

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Henry Ott — 'Electromagnetic Compatibility Engineering'
- ▶ Tim Williams — 'EMC for Product Designers'
- ▶ Würth Elektronik 'Trilogy of Magnetics' (free PDF)
- ▶ Robert Feranec / EMCFastPass YouTube channels
- ▶ BIS / WPC India compliance resources
- ▶ CISPR 32 / 35, EN 55032 / 55035 standards
- ▶ Schaffner / EM Test application notes
- ▶ Local EMC consultancies — BENT, RIE, Lakshmi Labs (India)

Thank you • Build something this week.

EMC Design Review Checklist

Use this before sending Gerbers to fab

- ▶ ✓ Ground plane is continuous (no splits)
- ▶ ✓ Every IC power pin has multi-value decoupling caps
- ▶ ✓ Every connector pin has TVS to chassis ground
- ▶ ✓ Every cable bundle has a common-mode choke
- ▶ ✓ Switching loops are small and direct
- ▶ ✓ Clock traces are short and routed near ground
- ▶ ✓ RF and SMPS sections shielded (or planned for)
- ▶ ✓ Connector shells bonded to chassis
- ▶ ✓ TVS pads can be replaced with 0 Ω for debug
- ▶ ✓ Test points on TVS lines for failure analysis
- ▶ ✓ Pre-compliance scan booked
- ▶ ✓ Lab schedule includes one re-test cycle